Embroidery Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Brother PE800 Embroidery Machine; Embroidery machine; Embroidery machine with stool and table

What it is used for

An embroidery machine stitches decorative designs, lettering, or patterns onto fabric using programmed instructions and thread. Correct hooping, design setup, and thread handling help produce accurate embroidery while protecting the fabric and machine.

Main parts

- Needle penetrates the fabric and forms stitches.
- Embroidery hoop holds the fabric taut during stitching.
- Thread spool and thread path supply thread to the needle.
- Bobbin supplies lower thread for stitch formation.
- Needle bar and presser foot guide needle movement and fabric control.
- Embroidery arm or carriage moves the hoop according to the design.
- Display or control panel selects designs and operating settings.
- Table and stool support the machine and operator where provided.

Before you start

1. Read the machine manual and identify the supported embroidery format and hoop size.
2. Inspect the needle, presser foot, hoop, thread path, bobbin, and power cord for damage.
3. Choose fabric, stabiliser, needle, and thread suitable for the design.
4. Transfer or select the approved design and check its size and placement.
5. Wind and install the bobbin and thread the machine according to the manual.
6. Place the machine on a stable table and keep the work area clear.

How to use

1. Switch off the machine before changing the needle, hoop, or other attachments.
2. Secure the fabric and stabiliser evenly in the hoop without stretching the fabric excessively.
3. Attach the hoop to the embroidery carriage and verify that it locks correctly.
4. Thread the upper thread and bobbin following the machine's threading path.
5. Select the design and position it using the display or approved controls.
6. Use the machine's design trace or positioning check if available to verify the stitching area.
7. Lower the presser foot as instructed and start stitching at the recommended speed.
8. Monitor thread tension, stitch quality, fabric movement, and machine sounds during operation.
9. Pause and switch off the machine before correcting tangles, thread breaks, or fabric misalignment.
10. After completion, remove the hoop, trim loose threads carefully, and inspect the finished design.

⚠ Safety

- Keep fingers away from the needle and moving embroidery carriage during operation.
- Switch off and unplug the machine before replacing a needle or performing maintenance.
- Do not operate the machine with damaged wiring, a loose needle, or a cracked hoop.
- Use the correct needle type and size for the fabric and thread.
- Keep loose clothing, jewellery, and hair away from moving parts.
- Do not pull or push the fabric while the machine is stitching.

Care & storage

- Remove lint and thread scraps from the bobbin area using the recommended method.
- Clean the needle plate and other accessible areas according to the manual.
- Replace bent or blunt needles and inspect hoops and thread guides regularly.
- Cover the machine when not in use and store threads in a clean, dry place.

Common mistakes to avoid

- Hooping fabric loosely or using an unsuitable stabiliser.
- Selecting a design larger than the installed hoop allows.
- Threading the upper thread or bobbin incorrectly.
- Pulling fabric or adjusting thread while the machine is stitching.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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